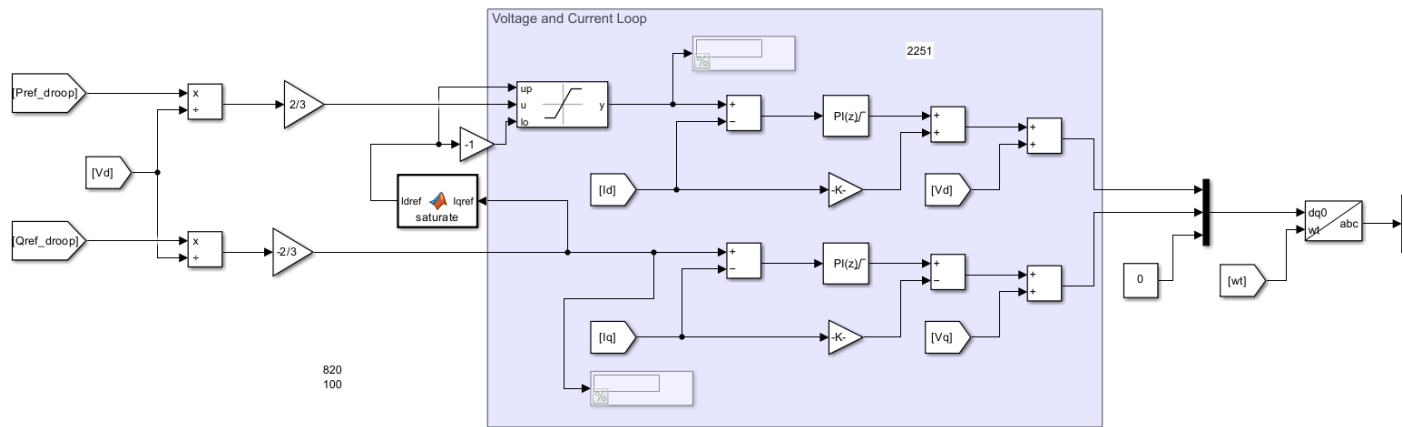


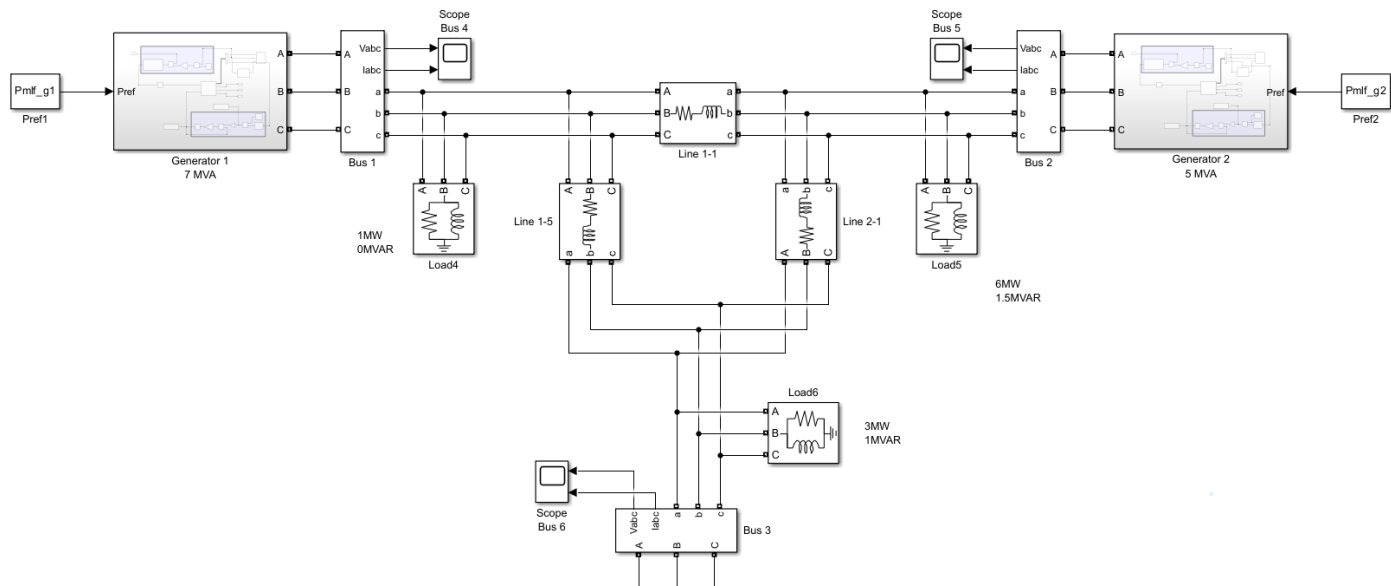
The diagram illustrates a power system architecture. It includes a PV Solar Array (labeled 'PV Solar Array') which is connected to a DC-DC Converter Unidirectional. This converter is connected to a DC-DC Converter Bidirectional, which is in turn connected to a battery. The battery is also connected to the DC-DC Converter Unidirectional. The system is controlled by a microcontroller (labeled 'u1') which outputs a duty cycle (D) to the Bidirectional converter. The battery's state of charge (SOC) and voltage are monitored. The load current (Iload) is measured. The DC link voltage (Vdclink) is monitored. The system is simulated in Simulink.

The diagram illustrates a three-phase inverter system. A DC input source (1) provides a 'dc In' signal to a 'Switching Bridge'. The bridge's output is connected to an 'i Measurement' block, which then feeds into the 'Inverter Controller'. The controller also receives a 'Pref' signal (1) and outputs a 'PWM' signal back to the bridge. The inverter's three-phase output (A, B, C) passes through an 'LCL Filter' (shaded blue area). This filter consists of two inductor blocks, 'Lf1' and 'Lf2', connected in series, with a capacitor block 'Cf' connected in parallel between their common nodes. The filtered output is then connected to a 'Vpp Measurement' block, which provides feedback to the 'Inverter Controller'. The final output is a three-phase signal (2) labeled 'a', 'b', and 'c'.

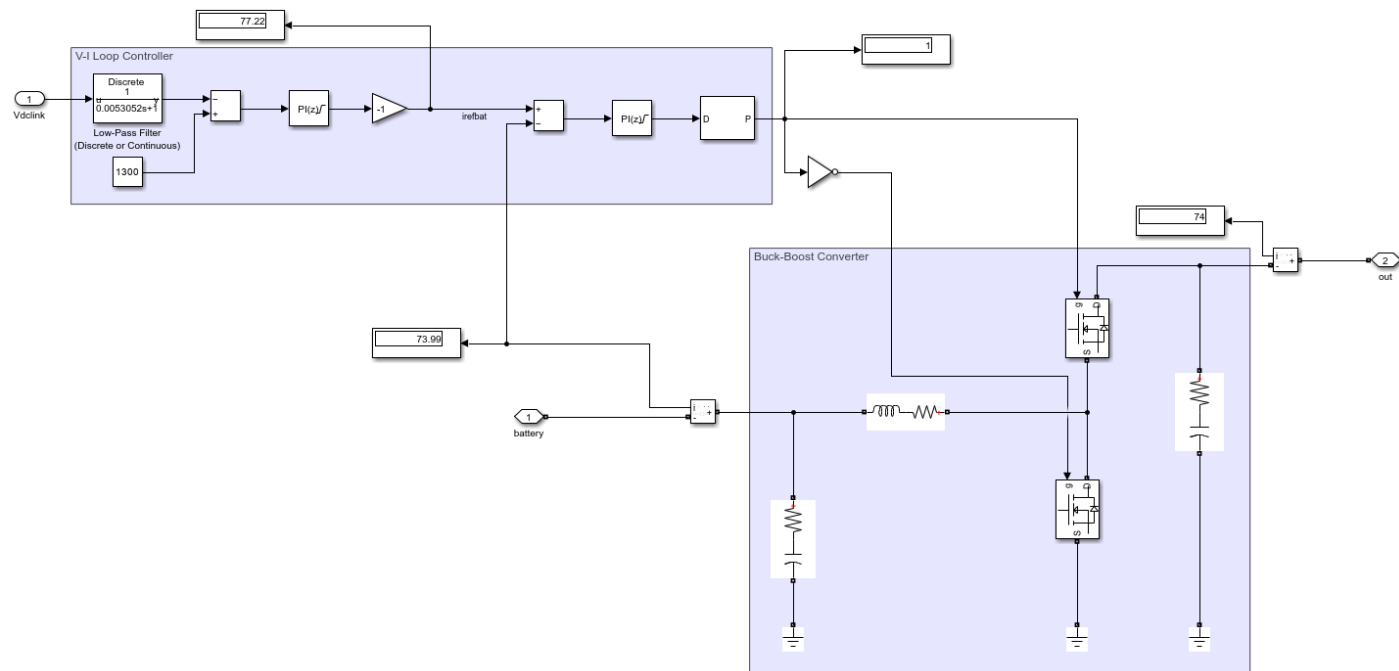
4. Inverter controller (2)



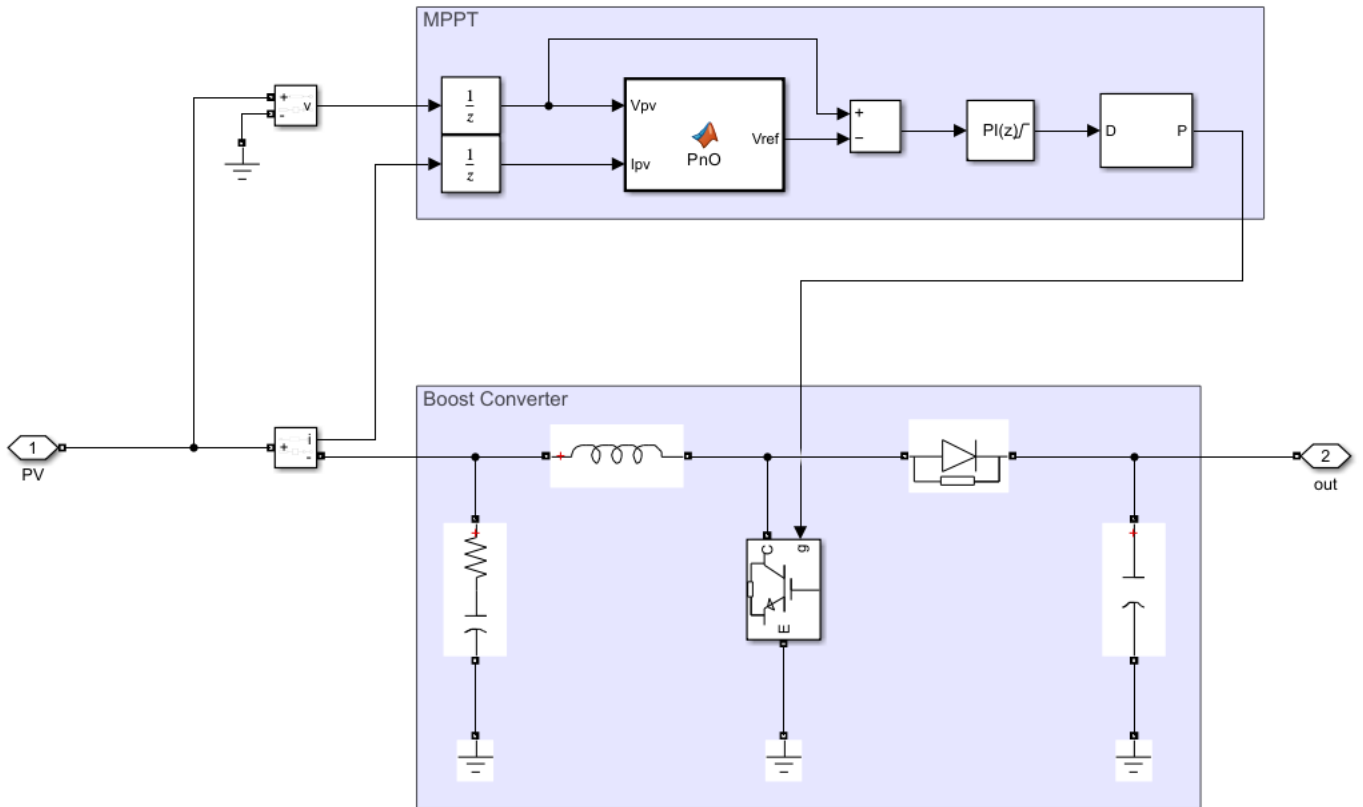
5. 3-bus grid



6. DC-DC battery



7. DC-DC PV



8. Generator system

